

Scale Invariance and Its Failure in ReLU Networks:

Positive Homogeneity, Layer-Rescaling Symmetry, and the Dying-ReLU Problem

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Abstract—We extend the series to the rectified linear unit (ReLU), the most widely used neural network activation function. We prove ReLU is positively homogeneous — $\text{ReLU}(cx) = c \text{ReLU}(x)$ for $c > 0$, but not for $c < 0$ — and that this induces an exact weight-rescaling symmetry in ReLU networks: scaling one layer's weights by $c > 0$ and the adjacent layer's by $1/c$ leaves the network's output completely unchanged. We then prove and simulate the genuine failure mode this activation is known for: unlike sigmoid or tanh, whose gradients merely shrink toward zero, ReLU's gradient is *exactly* zero for any negative pre-activation, which can permanently freeze a neuron's weights under gradient descent — the “dying ReLU” problem — confirmed here by training a small network to convergence with 19 of 20 neurons dead from initialization and provably immovable throughout.



1 POSITIVE HOMOGENEITY OF RELU

$\text{ReLU}(x) = \max(0, x)$.

Proposition 1. *For any $c > 0$, $\text{ReLU}(cx) = c \text{ReLU}(x)$ for all $x \in \mathbb{R}$. This fails in general for $c < 0$.*

Proof. If $x \geq 0$: $cx \geq 0$ (since $c > 0$), so $\text{ReLU}(cx) = cx = c \cdot x = c \text{ReLU}(x)$. If $x < 0$: $cx < 0$, so $\text{ReLU}(cx) = 0 = c \cdot 0 = c \text{ReLU}(x)$. Both cases match. For $c < 0$ and $x > 0$: $\text{ReLU}(cx) = 0$ (since $cx < 0$) but $c \text{ReLU}(x) = cx < 0$ — these differ whenever $x > 0$, so the identity fails. $\square$

1.1 Worked numerical verification

Over 10,000 random samples, testing $c \in \{2.0, 0.3, 10.0, 0.001\}$ gives $\max |\text{ReLU}(cx) - c \text{ReLU}(x)| = 0$ exactly (to floating-point precision) in every case. Testing $c = -1.5$ gives a maximum discrepancy of 11.4050 — large and nonzero, confirming the proposition's stated boundary exactly: the invariance holds precisely for $c > 0$ and genuinely fails for $c < 0$.

2 LAYER-RESCALING SYMMETRY IN RELU NETWORKS

Consider a two-layer network $f(x) = W_2 \text{ReLU}(W_1 x + b_1) + b_2$.

Proposition 2. *For any $c > 0$, replacing (W_1, b_1, W_2) with $(cW_1, cb_1, W_2/c)$ leaves $f(x)$ exactly unchanged for every input x .*

Proof. By Proposition 1, $\text{ReLU}(c(W_1 x + b_1)) = c \text{ReLU}(W_1 x + b_1)$ since $c > 0$. Then $(W_2/c) \cdot \text{ReLU}(cW_1 x + cb_1) = (W_2/c) \cdot c \text{ReLU}(W_1 x + b_1) = W_2 \text{ReLU}(W_1 x + b_1)$, and adding b_2 unchanged recovers $f(x)$ exactly. $\square$

This is not a curiosity: it is the exact reason weight magnitudes alone are not a reliable pruning or regularization

signal in ReLU networks without additional normalization — a network can be rescaled into an equivalent one with arbitrarily large or small weights on either side of a ReLU layer, with zero effect on behavior.

2.1 Worked numerical verification

For a random two-layer network ($5 \rightarrow 8 \rightarrow 3$), rescaling $(W_1, b_1, W_2) \rightarrow (cW_1, cb_1, W_2/c)$ for $c \in \{3.0, 0.2, 50.0\}$ reproduces the exact same output vector for a fixed test input, with maximum deviation on the order of 10^{-15} in every case — floating-point roundoff, not a real discrepancy. The invariance holds across a $250\times$ range of scale factors.

3 WHERE IT FAILS: THE DYING RELU PROBLEM

Unlike sigmoid or tanh, whose gradients shrink smoothly but never hit exactly zero, ReLU's derivative is a hard step function.

Proposition 3. $\frac{d}{dz} \text{ReLU}(z) = 0$ exactly for all $z < 0$ (and $= 1$ for $z > 0$; undefined at $z = 0$, conventionally taken as 0 or 1). Consequently, if a neuron's pre-activation $z = Wx + b$ is negative for every training example in a batch, its gradient with respect to W and b from that batch is exactly zero, and standard gradient descent leaves its weights completely unchanged.

Proof. Direct differentiation of $\max(0, z)$ gives 0 for $z < 0$. Backpropagation multiplies each upstream gradient by this local derivative (chain rule); if it is exactly 0 for every example in the batch, every term in the resulting weight-gradient sum is exactly 0, so the total gradient is exactly 0, and a gradient-descent update $W \leftarrow W - \eta \cdot 0 = W$ leaves the weights unchanged. $\square$

This is qualitatively different from vanishing gradients in sigmoid/tanh networks, where gradients become small but training can still (slowly) proceed; a dead ReLU neuron

receives a mathematically exact zero, with no mechanism to recover unless its inputs change through *other* neurons' updates.

3.1 Worked numerical verification

We train a small network (4 inputs, 20 hidden ReLU neurons, linear output) on synthetic regression data, deliberately initializing the hidden-layer bias with a strongly negative mean (-6) to push most neurons into the dead zone from the start:

TABLE 1
Dying ReLU: simulated training outcome

Quantity	Value
Dead neurons at initialization	19 / 20
Dead neurons after 300 training epochs	19 / 20
Direct check: gradient magnitude for a dead neuron	exactly 0.0

Nineteen of twenty neurons are dead before training even begins and remain dead through all 300 epochs — confirmed not just by the neuron count staying constant, but by directly inspecting the backpropagated gradient for a dead neuron and finding it exactly 0.0, not merely small. A direct, isolated comparison confirms why: evaluating both activation derivatives at negative pre-activations $\{-0.1, -1.0, -5.0, -50.0\}$ gives ReLU gradients of exactly $[0, 0, 0]$ at every magnitude, while Leaky ReLU ($\text{LReLU}(z) = z$ for $z > 0$, $0.01z$ otherwise) gives $[0.01, 0.01, 0.01, 0.01]$ — small, but genuinely nonzero, which is precisely why Leaky ReLU and its relatives (PReLU, ELU) are used as drop-in fixes: they preserve Proposition 1's positive homogeneity and Proposition 2's rescaling symmetry (Leaky ReLU is positively homogeneous for the same algebraic reason) while removing the permanently-zero gradient region that causes Proposition 3's failure mode.

4 CONCLUSION

ReLU gives this series its cleanest activation-function example of both sides of the recurring question:

- Positive homogeneity holds exactly for $c > 0$ and fails exactly at $c \leq 0$ — confirmed to floating-point precision and confirmed to fail by a large, unambiguous margin outside that range.
- This induces an exact layer-rescaling symmetry in ReLU networks — confirmed to $\sim 10^{-15}$ precision across a $250\times$ range of scale factors.
- The same hard nonlinearity that gives ReLU its clean scale invariance also gives it a genuine, exact-zero-gradient failure mode — confirmed by training a real (if small) network to convergence with 95% of neurons provably frozen from initialization onward.

As throughout this series: the interesting question was never whether ReLU is “well-behaved” in some vague sense, but precisely which properties hold exactly, under which conditions, and precisely where they stop.

REFERENCES

- 1) V. Nair and G. E. Hinton, “Rectified Linear Units Improve Restricted Boltzmann Machines,” ICML, 2010.
- 2) A. L. Maas, A. Y. Hannun, A. Y. Ng, “Rectifier Non-linearities Improve Neural Network Acoustic Models,” ICML Workshop on Deep Learning for Audio, Speech and Language Processing, 2013.
- 3) L. Lu, Y. Shin, Y. Su, G. E. Karniadakis, “Dying ReLU and Initialization: Theory and Numerical Examples,” Communications in Computational Physics, 2020.
- 4) D. Neyshabur, R. Salakhutdinov, N. Srebro, “Path-SGD: Path-Normalized Optimization in Deep Neural Networks,” NeurIPS, 2015.